

RESEARCH ARTICLE

Assessing experimental silvicultural treatments enhancing structural complexity in a central European forest – BEAST time-series analysis based on Sentinel-1 and Sentinel-2

Patrick Kacic¹ , Ursula Gessner² , Stefanie Holzwarth² , Frank Thonfeld²  & Claudia Kuenzer^{1,2}

¹Department of Remote Sensing, University of Würzburg, Institute of Geography and Geology, Am Hubland, Würzburg 97074, Germany

²German Aerospace Center (DLR), German Remote Sensing Data Center (DFD), Oberpfaffenhofen, Wessling 82234, Germany

Keywords

Beta-diversity, deadwood structures, forest management, forest structure, remote sensing

Correspondence

Patrick Kacic, Department of Remote Sensing, University of Würzburg, Institute of Geography and Geology, Am Hubland, 97074 Würzburg, Germany. Tel: +49 8153 284544; Fax: +49 (0)931 31-866690; E-mail: patrick.kacic@dlr.de

Funding Information

PK acknowledges funding from the DFG (Deutsche Forschungsgemeinschaft) within the framework of the Research Unit BETA-FOR (Enhancing the structural diversity between patches for improving multidiversity and multifunctionality in production forests) (grant no. FOR 5375/1, project number 459717468). FT acknowledges funding from the ForstEO project (FKZ 2220WK81A4), funded by the Federal Ministry of Food and Agriculture and Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection based on a decision of the German Bundestag from Waldklimafonds through the Fachagentur Nachwachsende Rohstoffe e.V. (FNR).

Editor: Prof. Mat Disney

Associate Editor: Dr. Wang Li

Received: 9 October 2023; Revised: 19 January 2024; Accepted: 13 February 2024

doi: 10.1002/rse2.386

Abstract

Assessing the dynamics of forest structure complexity is a critical task in times of global warming, biodiversity loss and increasing disturbances in order to ensure the resilience of forests. Recent studies on forest biodiversity and forest structure emphasize the essential functions of deadwood accumulation and diversification of light conditions for the enhancement of structural complexity. The implementation of an experimental patch-network in managed broad-leaved forests within Germany enables the standardized analysis of various aggregated and distributed treatments characterized through diverse deadwood and light structures. To monitor the dynamics of enhanced forest structure complexity as seasonal and trend components, dense time-series from high spatial resolution imagery of Sentinel-1 (Synthetic-Aperture Radar, SAR) and Sentinel-2 (multispectral) are analyzed in time-series decomposition models (BEAST, Bayesian Estimator of Abrupt change, Seasonal change and Trend). Based on several spatial statistics and a comprehensive catalog on spectral indices, metrics from Sentinel-1 ($n = 84$) and Sentinel-2 ($n = 903$) are calculated at patch-level. Metrics best identifying the treatment implementation event are assessed by change point dates and probability scores. Heterogeneity metrics of Sentinel-1 VH and Sentinel-2 NMDI (Normalized Multi-band Drought Index) capture the treatment implementation event most accurately, with clear advantages for the identification of aggregated treatments. In addition, aggregated structures of downed or no deadwood can be characterized, as well as more complex standing structures, such as snags or habitat trees. To conclude, dense time-series of complementary high spatial resolution sensors have the potential to assess various aggregated forest structure complexities, thus supporting the continuous monitoring of forest habitats and functioning over time.

Introduction

Recent climate change-induced drought conditions in central Europe have greatly impacted forest health (Buras et al., 2020; Rakovec et al., 2022; Schuldt et al., 2020). As a result, increasing canopy cover loss (Thonfeld et al., 2022) and tree mortality rates (Senf et al., 2020) have been reported due to decreasing climatic water balances, specifically in spring (Hajek et al., 2022). Research on future canopy mortality trajectories of Europe's forests classify recent forest mortality rates in excess of typical dynamics in the context of continuously increasing canopy mortality for the past three decades (Senf et al., 2021). Furthermore, the analysis of post-disturbance structures (1985 to 2018) highlights the significance of stand replacing disturbances and concludes that disturbance frequencies steadily increased, whereas disturbance severity presents declining trends (Senf & Seidl, 2021). Besides drought related disturbances, European temperate forests are exposed to an increased discrepancy between green and thermal seasons (Fu et al., 2023) which influences phenological cycles and productivity rates (Crabbe et al., 2016), thus requiring adaptive management.

Conflicting management interests for forests in Europe have been reported by various studies (Halme et al., 2010; Jakobsson et al., 2020; Krumm et al., 2023; Popkin, 2021; Schulze & Ammer, 2015). Those studies call for a more adaptive management reducing salvage logging, preserving areas undisturbed by logging and focusing on long-term management decision making that carefully evaluates economically driven practices against social and ecological impacts (Müller et al., 2018). Under the umbrella of the EU Biodiversity Strategy 2030 (European Commission, 2020a), the Nature Restoration Law (the first continent-wide law on nature restoration) targets an “increasing trend for standing and lying deadwood, uneven aged forests, forest connectivity, abundance of common forest birds and stock of organic carbon” (European Commission, 2023). In addition, the EU Forest Strategy builds on the EU Biodiversity Strategy contributing to greenhouse gas emission reductions through strengthening forest resilience and multifunctionality (European Commission, 2020b). To enhance forest biodiversity in Europe, multi-scale conservation efforts need to be implemented to reduce biodiversity loss threatening economic, social and ecological stability (Muys et al., 2022).

Forest management has multi-facet impacts on stand structure ranging from reduction in deadwood and microhabitats (Dieler et al., 2017) to overall decline in structural heterogeneity for a production-oriented management scenario (Li et al., 2023; Zeller et al., 2021). Chaudhary et al. (2016) conducted a global meta-analysis on species richness in the context of various management practices concluding that selection and retention systems or reduced impact logging harbor higher rates of local species richness than clear

cutting, agroforestry or plantations. Further studies by Paillet et al. (2010), Storch et al. (2023) and Zeller et al. (2023) highlight that diversification of age and species (old-growth structures), heterogeneity in structures at landscape level and continuous forest cover contribute to the highest species richness, thus supporting forest multifunctionality (van Der Plas et al., 2016a,b).

Nabuurs et al. (2022) stress that the integration of remote sensing for forest monitoring is essential to achieve the goals declared at the UN Climate Change Conference of the Parties (COP26). To characterize forest structure and its complexity at larger than landscape-level scale and short-term temporal intervals, spaceborne remote sensing offers great potentials due to the consistent acquisition of high spatial and temporal data from an increasing number of sensors (Camarretta et al., 2020; Huete, 2012; Jetz et al., 2016). In addition, the integration of remote sensing data for biodiversity conservation from an ecological and evolutionary perspective remains crucial for the assessment of ecosystem structure, composition and function in order to strengthen the link from laboratory or in-situ measurements to spaceborne observations (Cavender-Bares et al., 2022; Gonzalez et al., 2023; Skidmore et al., 2021). The sensor characteristics from Sentinel-1 (Synthetic-Aperture Radar, SAR) and Sentinel-2 (multispectral), namely a spatial resolution of up to 10 m and a theoretical temporal revisit time of about 5 days, allow for the generation of dense time-series to assess various land surface dynamics (Berger et al., 2012; Malenovsky et al., 2012). Recent reviews on forest monitoring in Germany by Holzwarth et al. (2020) and Holzwarth et al. (2023) list remote sensing studies making use of Sentinel-1 and/or Sentinel-2 data for disturbance mapping (Kaiser et al., 2022; König et al., 2023; Scharvogel et al., 2020; Thonfeld et al., 2022; Zimmermann & Hoffmann, 2020), the assessment of forest structure dynamics (Kacic et al., 2023) and biodiversity monitoring (Bae et al., 2019; Hoffmann et al., 2022; Liu et al., 2023). Nevertheless, an in-depth assessment of remote sensing time-series for the characterization of various forest structure complexities remains understudied.

To better understand the influence of different silvicultural treatments for enhancing structural complexity and beta-diversity in German broad-leaved production forests, an experimental patch-network was established in the context of the trans-disciplinary BETA-FOR project (Mueller, Mitesser, et al., 2022; Mueller, Scherer-Lorenzen, et al., 2022). The patches ($n = 234$) cover an area of 50×50 m and are distributed over six regions in Germany: forest of the Julius-Maximilians-University of Würzburg (“University Forest”), Bavarian Forest national park, church forest of the diocese Passau, Huensrück-Hochwald national park and the municipality forests of Saarland and Lübeck. A wide range of

standardized disturbance treatments, subdivided into aggregated and distributed treatments, promote the diversification of deadwood and light structures (Fig. 1).

The experimental set-up mimics a great variety of forest structures that can on the one hand result from natural disturbance (e.g. windthrow), and on the other hand from silvicultural management (e.g. selective thinning, partial cutting). The wide-range of established treatments has very different ecological implications for available

habitats and resources, thus altering biodiversity (Messier et al., 2021; Mueller, Mitesser, et al., 2022; Rothacher et al., 2023). To the best of our knowledge, the patch-network established in the context of the BETA-FOR project is the first large-scale simulation experiment on structural beta-diversity in temperate broad-leaved forests. Therefore, the BETA-FOR project addresses research gaps, such as the need for structural diversification (Messier et al., 2021), microclimate

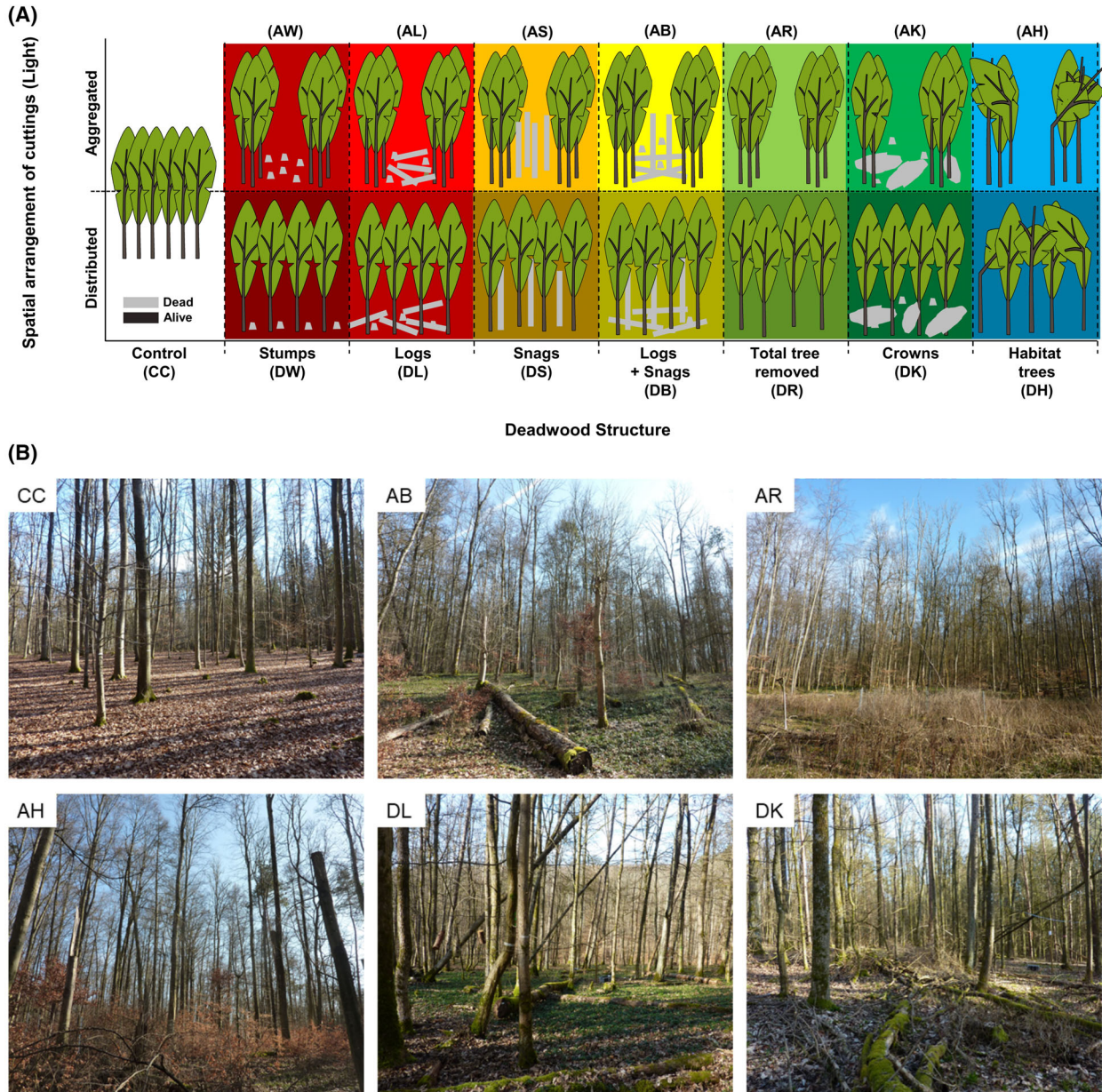


Figure 1. Schematic representation of the BETA-FOR treatments (A) characterized by varying deadwood structure and spatial arrangement of cuttings (adapted from Mueller, Mitesser, et al. (2022)). The photos (B) show a selection of treatments at the BETA-FOR region "University Forest" in March 2023 (photos taken by P. Kacic).

measurements in varying forest structures (Großmann et al., 2023; Zellweger et al., 2019, 2020), deadwood manipulation experiments (Sandström et al., 2019), influence of light availability on plant species richness (Dormann et al., 2020) and structural complexity in the context of multi-trophic groups (Penone et al., 2018). From a remote sensing perspective, the experimental patch-network of aggregated and distributed forest structure complexities enables a validated assessment of remote sensing time-series in order to fill the research gaps (1) to which extent the different treatment implementations can be identified as change points in a time-series, (2) if there are different capabilities to assess aggregated and distributed treatments and (3) how the sensor type influences the characterization of forest structure complexity.

To address these research gaps, satellite time-series of Sentinel-1 and Sentinel-2 are examined to assess the pre-disturbance, implementation (disturbance) and post-disturbance conditions of BETA-FOR treatments at patch-level. Various time-series algorithms have been developed in recent years to assess seasonal and trend dynamics: The early study by Verbesselt et al. (2010) proposed the BFAST algorithm detecting seasonal and trend changes in earth observation time-series through the decomposition of an original time-series into seasonal, trend and residual components. The publication by Kennedy et al. (2010) presented the LandTrendr algorithm based on annual Landsat time-series to analyze trends in forests (disturbance, recovery) through temporal segmentation. Zhu and Woodcock (2014) also developed a time-series method based on Landsat data for the continuous change detection and classification of land cover (CCDC). The Jumps Upon Spectrum and Trend (JUST) model was published by Ghaderpour and Vujadinovic (2020) focusing on a spectral analysis (ALLSSA, Anti-Leakage Least-Squares Spectral Analysis) to identify changes in time-series of vegetation indices. The first Bayesian time-series decomposition algorithm, namely Bayesian Estimator of Abrupt change, Seasonal change and Trend models (BEAST), was developed by Zhao et al. (2019) and comes with several advantages: (1) combination of multiple models (i.e. no criterion-based algorithm resulting in a single best model, e.g. BFAST); (2) due to the probabilistic nature, a validity (change point probability) is assessed to delineate different likelihoods of change; (3) no user-defined specifications are needed (e.g. definition of a training and monitoring period) and (4) the generic algorithm can be applied to any time-series (regular and irregular). In addition, Zhao et al. (2019) and other studies (e.g. Cai et al. (2020); Li et al. (2022)) demonstrated an increased performance and accuracy of BEAST compared to BFAST based on various time-series data which is why the BEAST algorithm is the

choice of the present study. BEAST models are tested with various metrics (combination of spectral indices and spatial statistics) derived from Sentinel-1 and Sentinel-2 to estimate the seasonality and trend components and change point probabilities. Spectral indices from Sentinel-1 and Sentinel-2 data are calculated based on a comprehensive catalog (Montero et al., 2023) aiming to propose spectral indices and spatial statistics (aggregation of pixels in BETA-FOR treatments, i.e. patches) best identifying the treatment implementation according to change point probability and date. The following hypotheses are investigated in the present study:

- Satellite-derived metrics from Sentinel-1 and Sentinel-2 can assess the implementation of treatments to a sensor-specific extent. Dense time-series and high spatial resolution allow for a continuous monitoring of forest structure conditions, both being key criteria to identify the treatment implementation.
- The identification of treatment implementations according to BEAST change point probability and date might be more accurate for aggregated treatments than distributed treatments since aggregated treatments result in clear spectral signals at 10–20 m pixel-level, whereas distributed treatments can only be captured as a mixed-pixel signal.

Materials and Methods

Study area

The present study focuses on the BETA-FOR region “University Forest” in central Germany belonging to the Julius-Maximilians-University of Würzburg which is an intensively managed broad-leaved forest and well-known hotspot for biodiversity research (Bae et al., 2019; Heidrich et al., 2020; Mueller, Scherer-Lorenzen, et al., 2022). The mixed forests are dominated by European beech (*Fagus sylvatica*) on calcareous soils (Mueller, Scherer-Lorenzen, et al., 2022). At this region, 60 patches of the BETA-FOR research network are located (Fig. 2).

Each treatment type (Fig. 1A) was established three times except “DW” (distributed stumps) serving as specific reference for selective thinning. The “University Forest” patches were established during November and December 2018, while the five other BETA-FOR regions were implemented in winter 2015/2016 or winter 2016/2017. Because of the treatment implementation in early winter 2018 at the “University Forest” region, the present study is limited to this region, since a multi-annual pre-disturbance time-series of Sentinel-1 and Sentinel-2 data is available, thus allowing for a valid time-series decomposition and change point evaluation. Furthermore, the “University Forest” region is the primary focus of the

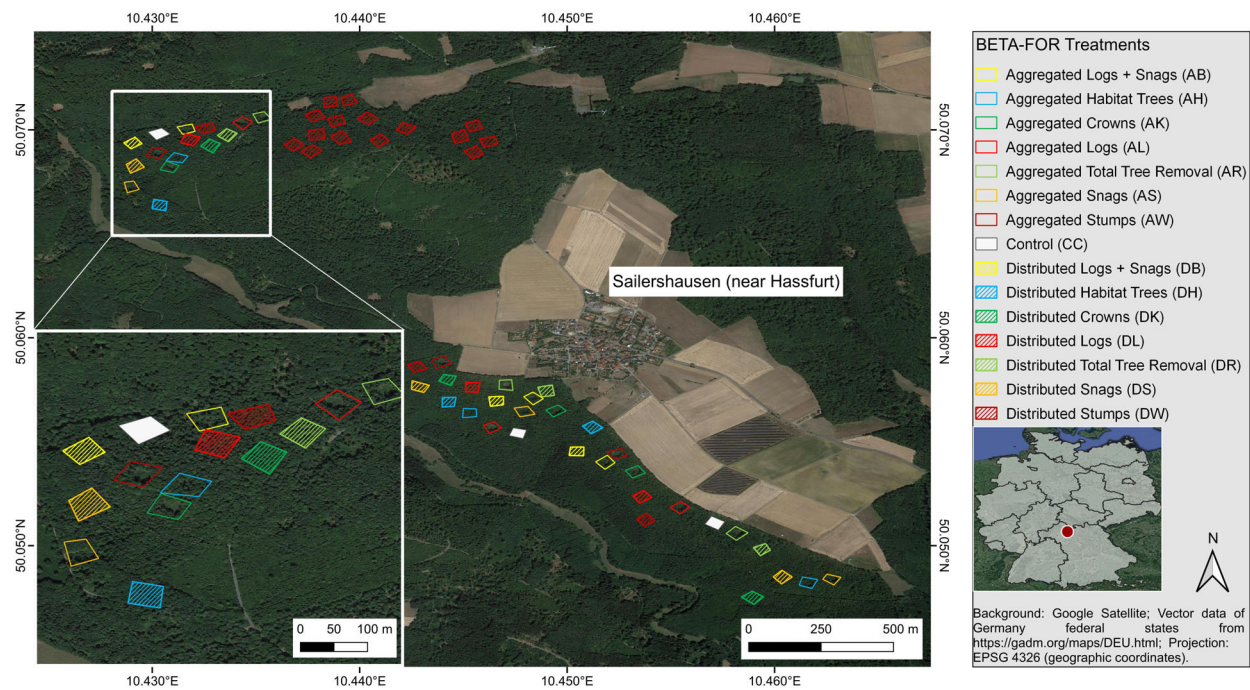


Figure 2. Map of the study area in the forest of the University of Würzburg near Sailershausen in central Germany. The different treatment patches are grouped into control, aggregated and distributed, colorized according to Figure 1A.

BETA-FOR project since six additional treatment types (AR, DR, AK, DK, AH, DH; the explanation of abbreviations is present in Fig. 1) are established in this forest region. Each patch covers an area of 50 × 50 m where the forest structure complexity of distributed treatments was manipulated in the whole patch area in contrast to the aggregated treatments (centered area with 30 m diameter, i.e. no manipulation in the outer patch-area). In both aggregated and distributed treatments, about 30% of the patch tree biomass was manipulated. In addition, control patches were implemented serving as a reference of unaltered forest structure conditions. More information on the forest structure composition of the “University Forest” is found in Mueller, Scherer-Lorenzen, et al. (2022).

Sentinel-1 and Sentinel-2 data

To obtain high spatial and temporal observations that sufficiently cover both the pre- and post-disturbance conditions at the BETA-FOR patches in the “University Forest”, data from Sentinel-1 and Sentinel-2 was examined (Berger et al., 2012; Malenovsky et al., 2012). In the present approach, we made use of the full Sentinel-1 and Sentinel-2 time-series archive available (Table 1). Both Sentinel-1 and Sentinel-2 consist of a pairwise satellite constellation (end of mission of Sentinel-1B since December 2021), thus enabling high revisit times (minimum of

Table 1. Sentinel-1 and Sentinel-2 characteristics: sensor types, data levels, temporal periods, bands.

Sensor name	Sentinel-1	Sentinel-2
Sensor type	Synthetic-Aperture-Radar (SAR)	Multispectral
Data level	Ground-Range-Detected (GRD)	Surface Reflectance (L2A)
Temporal period	10–2014 to 10–2022	07–2015 to 10–2022
Mean observations among all BETA-FOR patches in “University Forest” after pre-processing (10–2014 to 10–2022)	754 observations	248 observations
Mean observations among all BETA-FOR patches in “University Forest” after pre-processing in period of treatment implementation (11–2018 to incl. 12–2018)	20 observations	4 observations
Bands	VV, VH (both 10 m)	VIS, NIR, SWIR (all 10 and 20 m bands available)

about 4 days in central Germany). Sentinel-1 Ground-Range-Detected (GRD) data from October 2014 to October 2022 was processed via the Google Earth Engine

(GEE, Gorelick et al. (2017)) by applying the analysis-ready processing workflow (border noise removal, speckle filtering, radiometric terrain normalization) by Mullissa et al. (2021). Sentinel-2 L1C data (July 2015 to March 2017) was obtained and processed through cluster-processing infrastructure of the German Remote Sensing Data Center (DFD) of the German Aerospace Center (DLR) to L2A surface reflectance using Sen2Cor (Main-Knorn et al., 2017), since this initial period of Sentinel-2A data is not archived as L2A in GEE. Sentinel-2 L2A surface reflectance imagery spanning the period from April 2017 to October 2022 was accessed through GEE (Sen2Cor processing (Main-Knorn et al., 2017)). Therefore, a consistent time-series of Sentinel-2 L2A data spanning the period from July 2015 to October 2022 was prepared for further analysis. For Sentinel-1, the polarization bands VV and VH (both 10 m) were selected, whereas for Sentinel-2 all 10 m and 20 m bands were used. The Sentinel-2 data was cloud-masked based on the scene classification (Main-Knorn et al., 2017) and the cloud probability layer (Google Earth Engine Developers, 2022) in order to remove atmospheric artifacts. The difference in number of high-quality observations of Sentinel-1 and Sentinel-2 is primarily resulting from the insensitivity towards atmospheric conditions of Sentinel-1. Therefore, about three times more Sentinel-1 observations are available when considering the full time-series and five times more observations when focusing on the treatment implementation period (November and December 2018, Table 1).

Indices calculation and spatial statistics

The calculation of spectral indices from various remote sensing sensor types for vegetation monitoring has been demonstrated in a great number of studies (Huete, 2012; Huete et al., 2002; Rouse et al., 1974; Teillet et al., 1997; Xue & Su, 2017; Zeng et al., 2022). Spectral indices offer the benefits that spectral signatures of objects can be identified, confounding effects are removed or weakened and targeted monitoring of certain phenomena are possible. Also because of the availability of three red-edge bands capturing the spectrally sensitive wavelength range of chlorophyll content, Sentinel-2 has great capabilities for vegetation monitoring (Bhattarai et al., 2020; Delegido et al., 2011; Frampton et al., 2013; Sun et al., 2020). Recent studies on German forests based on Sentinel-1 VV and VH polarization have highlighted the potential of C-band radar for the detection of drought affected forest stands (Kaiser et al., 2022), modeling of forest structure characteristics (Kacic et al., 2023) and biodiversity monitoring (Bae et al., 2019; Hoffmann et al., 2022; Liu et al., 2023).

Based on a standardized and comprehensive catalog of spectral indices (Montero et al., 2023), all listed spectral indices were calculated after filtering by “application_domain” (“radar” for Sentinel-1; “vegetation” and “water” for Sentinel-2; a list of considered spectral indices is found in Data S1). Water indices were included since e.g. the Normalized-Difference-Water-Index (NDWI) is sensitive to vegetation liquid water (Gao, 1996). At the date of accessing the spectral indices catalog (22th of September 2022), spectral indices on soil characteristics were still classified under “vegetation” according to “application_domain.” Therefore, ten different radar indices and 129 multispectral indices were calculated allowing for a consistent assessment of remote sensing indices time-series for the identification of BETA-FOR treatment implementations. For Sentinel-1, the ten radar indices were complemented by VV and VH polarization, thus resulting in 12 bands/indices in total. For each observation of the Sentinel-1 and Sentinel-2 time-series, the pixels of each remote sensing index were aggregated as spatial statistics ($n = 7$) at patch-level (clipped to vector polygon of patches) to assess extreme (minimum, maximum), average (mean, median) and heterogeneous conditions (standard deviation, variance, coefficient of variation). As a result, for each BETA-FOR patch ($n = 60$), the total number of metrics (combination of spectral index and spatial statistic) for Sentinel-1 and Sentinel-2 amounts to 84 and 903, respectively. The workflow of data processing is also depicted in Figure 3.

BEAST time-series analysis

To assess the implementation event and temporal dynamics in the BETA-FOR patches, the BEAST algorithm (Zhao et al., 2019) was used in order to decompose the time-series of remote sensing metrics from Sentinel-1 and Sentinel-2 into trend and seasonal components and assess change points. The potential of the probabilistic approach of BEAST, integrating Bayesian model averaging through ensemble learning to model complex natural dynamics which cannot be accounted for by a single-model, has been demonstrated in Zhao et al. (2019). Above all, the calculation of change point probabilities supports the interpretation of gradual and abrupt changes and their validity. Therefore, the susceptibility to false positives is reduced (Awty-Carroll et al., 2019; Ghaderpour & Vujadinovic, 2020) compared to other single-best-model algorithms (e.g. BFAST, Verbesselt et al. (2010)). Furthermore, trend and seasonal change points are assessed separately since there does not necessarily need to be a temporal overlap. The BEAST workflow can be subdivided into the following processing steps: (1) the data input to the BEAST model is a time-series that captures the seasonal, abrupt and trend

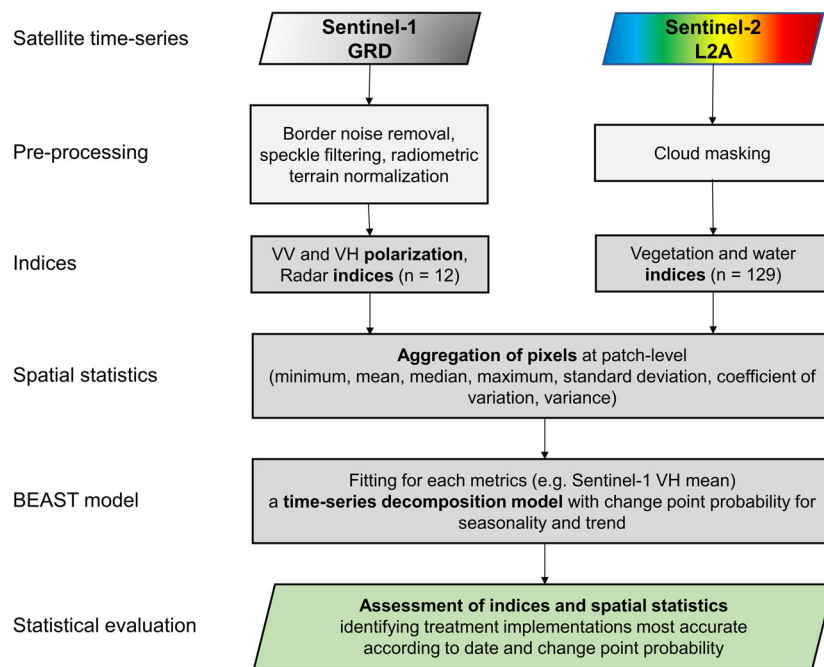


Figure 3. Workflow chart of Sentinel-1 and Sentinel-2 processing for time-series decomposition and statistical assessment of indices and spatial statistics identifying BETA-FOR treatment implementations.

dynamics of the observed system (e.g. a time-series on forest structure complexity); (2) multiple competing models are calculated to decompose the original time-series into additive seasonal and trend components; (3) under the assumption that there is no single best model to fit a time-series, all models are considered useful to some extent and are combined (i.e. ensemble learning instead of choosing one model) based on Bayesian model averaging and (4) the calculation of a weighted average model through Bayesian model averaging allows the quantification of change point probabilities for both seasonal and trend change points. A graphical illustration of the BEAST workflow can be found in the publication of Zhao et al. (2019), Fig. 2. In the present study, we apply the BEAST algorithm in order to identify remote sensing metrics from Sentinel-1 and Sentinel-2 that assess the treatment implementation as high probability change points in trend and/or seasonality. More specifically, we defined three criteria that need to be fulfilled so that a metric is classified as a proxy for the valid identification of treatment implementation: (1) a change point needs to be detected within the period of treatment implementation (November and December 2018), (2) reaching a change point probability of >90% (threshold defined after testing for high probability change points) and (3) holding the maximum change point probability of all change points assessed in the time-series for the respective metric. The third criteria was defined because we consider the treatment implementations as strongest disturbance dynamic

(exceeding natural dynamics) within the period of the time-series since we did not find any information on major disturbance events (windthrow, insect infestation, fire). Therefore, we run a time-series decomposition model for each remote sensing metric calculated from Sentinel-1 and Sentinel-2 at each BETA-FOR patch. In the following, we assessed change points that identify the treatment implementation and analyzed pre- and post-disturbance characteristics. Statistical evaluations were made to highlight remote sensing indices and spatial statistics that best determine accurate change points identifying the treatment implementations. In addition, comparative statistics of seasonal and trend change points were carried out.

Results

General assessment of indices and metrics

The following results show which spectral indices and metrics resulted in a consistent detection of change points. A detection is considered consistent if spectral indices and metrics assessed accurate change points (according to our validity criteria stated in the previous subsection) for all individual patches of a certain treatment. Furthermore, spectral indices are listed that did not identify any accurate change point, independent of the treatment type. Results of this subsection present pooled statistics of seasonal and trend change points.

Table 2. Sentinel-1 and Sentinel-2 spectral indices that identify consistently (i.e. in all individual patches of certain treatments) accurate change points.

BETA-FOR treatment	Sentinel-1	Sentinel-2
AB	VH	AVI, AWEIsh, FCVI, GEMI, MBWI, MLSWI26, MSI, NMDI, TriVI
AH	–	NMDI
AK	VVVHS	AWEIsh, MBWI, MNDWI, MSI, NDMI, NDPI, SR3
AL	–	MBWI, NDPI, NDREI, SARVI
AR	–	MSI, NDREI, NMDI
AS	DPDD, VHVVP, VVVHS	–
AW	VH	CIRE, DVI, GOSAVI, LSWI, MSI, MTVI1, NDII, NDMI, NDPI, NDREI, NIRv, NMDI, WDVl

The long names of spectral indices are present in Data S1.

For both Sentinel-1 and Sentinel-2, only aggregated treatments could be identified consistently using remote sensing indices, independent of the spatial statistics (Table 2). Furthermore, there is no single-best index that assesses all aggregated treatments, either for Sentinel-1 or Sentinel-2. For Sentinel-1, metrics based on VH polarization or on VVVHS (VV and VH sum) index could each identify two treatments consistently, i.e. AB, AW and AK, AS, respectively. Sentinel-2 indices consistently assessing at least three treatment types are NMDI (Normalized Multi-band Drought Index) and NDREI (Normalized Difference Red Edge Index). Aggregated treatments that could not be determined consistently are AH, AL and AR for Sentinel-1, whereas Sentinel-2 spectral indices could assess all aggregated treatments except AS.

The consistency analysis of Sentinel-1 and Sentinel-2 metrics highlights that spectral indices combined with spatial heterogeneity statistics assess most treatments (Table 3). For Sentinel-1, VVVHS coefficient of variation could identify consistently the implementation event as change point for all individual treatments in AK and AS. On the one hand, Sentinel-2 derived NMDI coefficient of variation accurately assessed the treatment implementation for AR and AW, and on the other hand, NDPI (Normalized Difference Phenology Index) variance assessed both AK and AW. In addition, the standard deviation of NDREI identified accurate change points for the treatments AL and AW.

In Table 4, spectral indices are listed that did not detect any treatment implementation correctly, independent of the treatment type and spatial statistic. About 42% ($n = 5$) of all Sentinel-1 bands/spectral indices, and about 19% ($n = 25$) of all Sentinel-2 spectral indices did not

Table 3. Sentinel-1 and Sentinel-2 metrics (combination of index and spatial statistics) that identify consistently (i.e. in all individual patches of certain treatments) accurate change points.

BETA-FOR treatment	Sentinel-1	Sentinel-2
AB	–	MBWI cv, MLSWI26 variance
AH	–	NMDI variance
AK	VVVHS cv	NDPI variance, SR3 cv
AL	–	NDPI std, NDREI std, MBWI cv
AR	–	NDREI variance, NMDI cv
AS	DPDD cv, VHVVP cv, VVVHS cv	–
AW	VH cv	CIRE std, LSWI std, LSWI variance, NDII std, NDMI std, NDMI variance, NDREI std, NDPI variance, NMDI cv, MSI cv, MSI std.

Abbreviations of spatial statistics: cv, coefficient of variation; std, standard deviation. The long names of spectral indices are present in Data S1.

Table 4. Sentinel-1 and Sentinel-2 indices not assessing any accurate change point identifying BETA-FOR treatment implementations.

Sentinel-1	Sentinel-2
$n = 5$ (42%) DpRVIHH, QpRVI, RFDI, VDDPI, VHVVR	$n = 25$ (19%) BCC, GARI, GLI, GRNDVI, IKAW, IPVl, MCARI705, MGRVI, MNLI, MSR, NDVI, NDYI, NWI, RCC, REDSI, RGBVI, SAVI2, SR, SR2, SWM, TCARIOSAVI705, TTVI, VIG, WDRVI, WRI

The long names of spectral indices are present in Data S1.

identify any accurate change point in the context of treatment implementation. Interestingly, the well-established multispectral indices NDVI (Normalized Difference Vegetation Index) and SR (Simple Ratio of near-infrared and red band) are among these spectral indices without successful detection.

Exemplary time-series visualization

To better understand the temporal characteristics of Sentinel-1 and Sentinel-2 metrics identifying the treatment implementation events, time-series of Sentinel-1 VH coefficient of variation (Fig. 4) and Sentinel-2 NMDI coefficient of variation (Fig. 5) are compared for three exemplary treatments. The original time-series of control (CC), aggregated (AW) and distributed treatments (DR) are decomposed into additive seasonal and trend components. It is important to note, that the respective treatment patches per treatment type are identical for both Sentinel-1 and Sentinel-2, thus allowing a comparison.

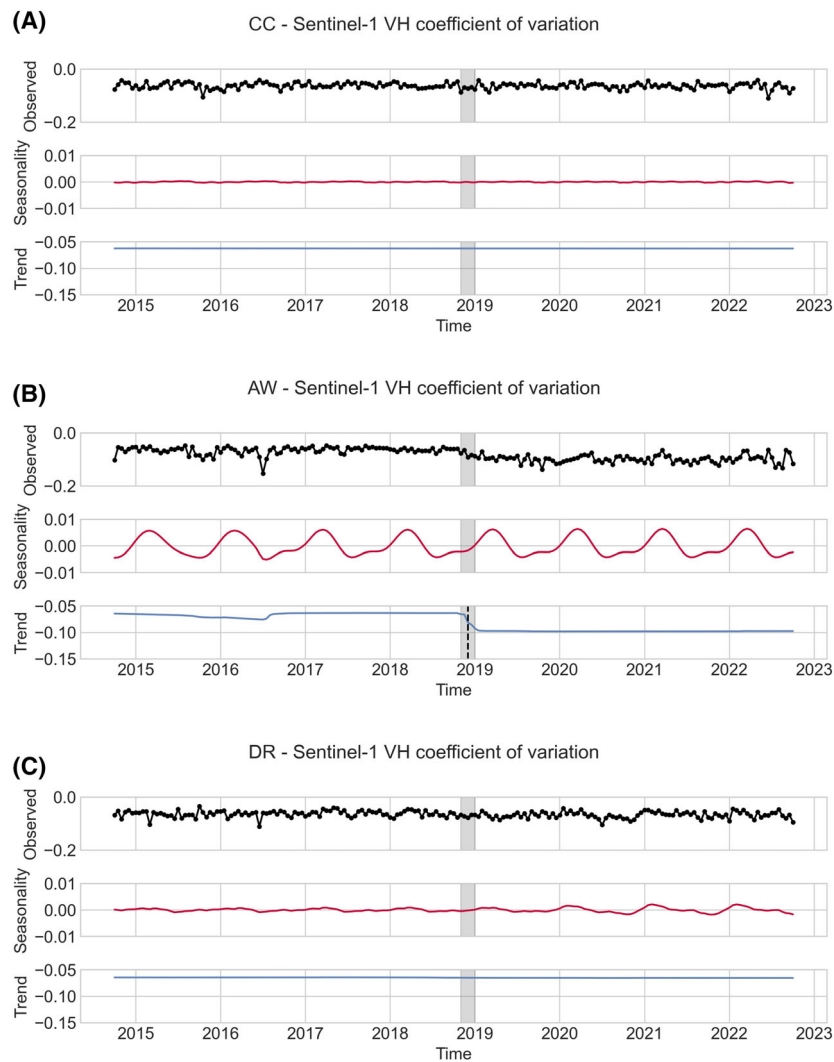


Figure 4. Exemplary Sentinel-1 VH coefficient of variation time-series for different BETA-FOR treatment patches: control (A), aggregated stumps (B) and distributed removal (C). Change points with a high probability (>90%) are depicted as a dashed vertical line. The gray area depicts the period of treatment implementation (disturbance). The respective treatment patches are identical with the Sentinel-2 time-series visualization (Fig. 5).

The Sentinel-1 time-series presents stable conditions for the control treatment (Fig. 4A) indicating no disturbance event, seasonal or trend signal. In comparison, the aggregated stumps treatment (AW, Fig. 4B) presents a strong shift in the trend at the time of treatment implementation which was identified as change point. For the distributed removal of complete trees including stumps (DR, Fig. 4C), no change points were detected although the seasonal component indicates an increasing amplitude with a temporal lag-effect of about 1 year with respect to treatment implementation.

Similar to the Sentinel-1 time-series, the exemplary time-series of Sentinel-2 NMDI coefficient of variation highlight the accurate assessment of the treatment implementation event for the aggregated treatment (Fig. 5B).

In comparison, the treatment implementation was clearly identified as change points both in the seasonal and trend component. At the treatment implementation event, the trend presents a strong increase and post-disturbance characteristics in the seasonal component are more extreme (higher annual amplitude). For both the control (Fig. 5A) and distributed treatment (Fig. 5C), no change points have been identified due to the relatively stable conditions without abrupt changes.

Statistical analysis

Spectral indices

The comparative statistics of Sentinel-1 and Sentinel-2 derived spectral indices that accurately assessed change

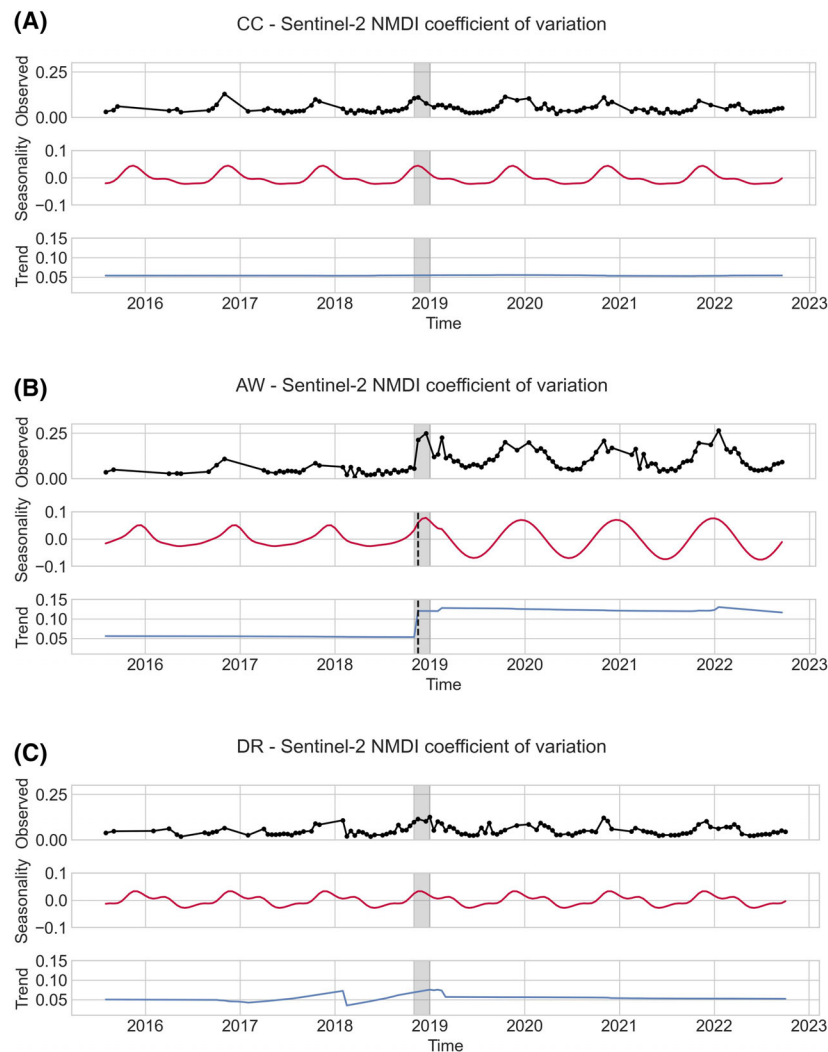


Figure 5. Exemplary Sentinel-2 NMDI coefficient of variation time-series for different BETA-FOR treatment patches: control (A), aggregated stumps (B) and distributed removal (C). Change points with a high probability ($>90\%$) are depicted as a dashed vertical line. The gray area depicts the period of treatment implementation (disturbance). The respective treatment patches are identical with the Sentinel-1 time-series visualization (Fig. 4).

points for the treatment implementation event present different capabilities according to sensor type and spectral indices (Fig. 6). Percentage values on the x-axis are calculated per spectral index accounting for all treatment patches ($n = 60$), change point types ($n = 2$) and spatial statistics ($n = 7$), i.e. considering all theoretically possible change points. Therefore, low percentage values are no surprise since the accurate assessment of the treatment implementation is influenced by treatment type, spectral index and spatial statistic (Tables 2–4). For Sentinel-1, the VH polarization assesses change points most accurately in comparison to all other Sentinel-1 bands/spectral indices (Fig. 6A). The spectral indices statistics for Sentinel-2 present the NMDI as best indicator identifying

about two times for accurate change points than Sentinel-1 VH. Furthermore, Sentinel-2 spectral indices reach higher percentage values compared to Sentinel-1 bands/spectral indices.

Spatial statistics

Similar to the statistics of spectral indices, the comparative analysis of spatial statistics for the identification of treatment implementation events shows differences between sensors, although the coefficient of variation performs best for both sensors (Fig. 7). For Sentinel-1, there is a clear advantage of using the coefficient of variation compared to the other spatial statistics since about four

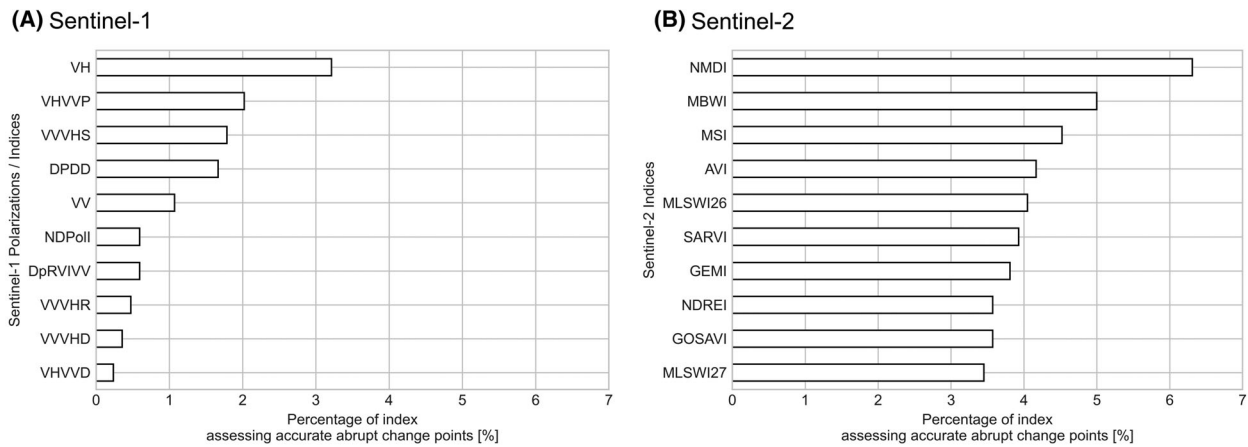


Figure 6. Comparative statistics of Sentinel-1 (A) and Sentinel-2 (B) indices assessing accurate change points depicting ten spectral indices each best assessing the treatment implementation event. Percentage values on the x-axis are calculated per spectral index accounting for theoretically possible change points for all treatment patches ($n = 60$), change point types ($n = 2$) and spatial statistics ($n = 7$).

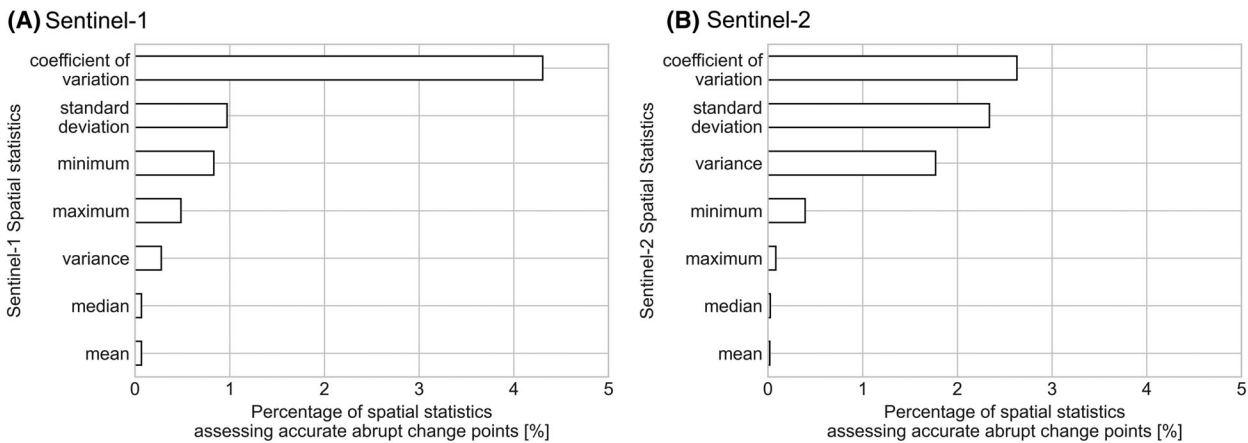


Figure 7. Comparative statistics of Sentinel-1 (A) and Sentinel-2 (B) spatial statistics assessing accurate change points. Percentage values on the x-axis are calculated per spatial statistic accounting for all treatment patches ($n = 60$), change point types ($n = 2$) and bands/spectral indices (Sentinel-1: $n = 12$, Sentinel-2: $n = 129$).

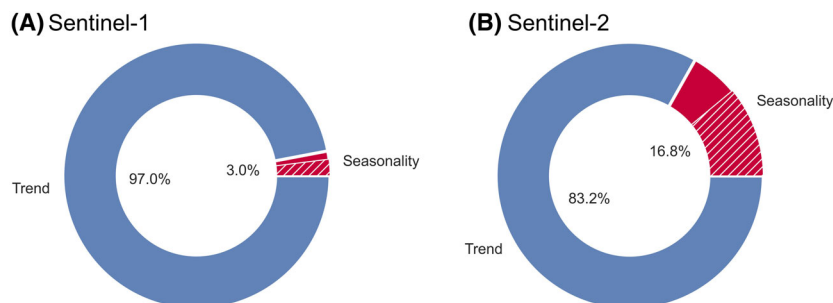


Figure 8. Comparative statistics of Sentinel-1 (A) and Sentinel-2 (B) accurately assessed change points. The hashed areas represent the temporal overlap of trend and seasonality change points, i.e. seasonality change points that are also assessed as trend change points (Sentinel-1: 66.6%, Sentinel-2: 65.6%).

times more accurate change points for the treatment identification are assessed (Fig. 7A). Sentinel-2 derived statistics highlight that heterogeneity statistics outperform extreme and average statistics, although the coefficient of variation reaches similar percentage values as standard deviation and variance statistics (Fig. 7B). Again, low percentage values are a result of the calculation taking all theoretically possible change points into account, i.e. from different treatments ($n = 60$), change point types ($n = 2$) and bands/spectral indices (Sentinel-1: $n = 12$, Sentinel-2: $n = 129$).

Change points

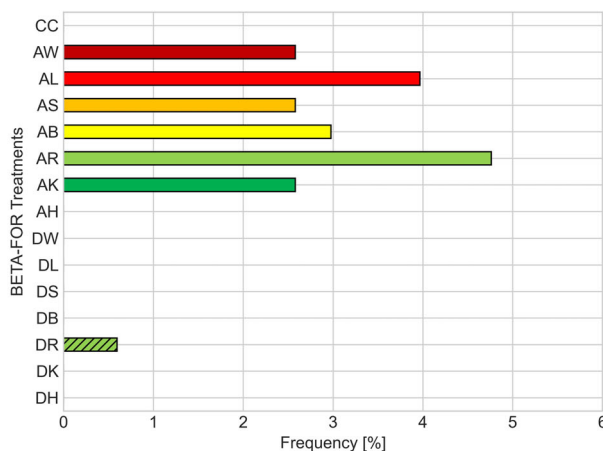
The comparative statistics of the change point types “seasonality” and “trend” emphasize for both Sentinel-1 and Sentinel-2 that the great majority of accurately assessed change points are detected in the trend signal (Sentinel-1: 97.0%, Sentinel-2: 83.2%, Fig. 8). When considering the temporal overlap of change points detected in the seasonal and trend components, for both sensors about two thirds of all seasonal change points are also assessed as trend change points (Sentinel-1: 66.6%, Sentinel-2: 65.6%).

BETA-FOR treatments

The following comparative statistics on the capability of Sentinel-1 and Sentinel-2 metrics to assess the implementation of various BETA-FOR treatments incorporate previous results that there is a varying consistency among spectral indices and spatial statistics, and that most

change points are accurately assessed in the trend component. Those results need to be taken into account when evaluating percentage values describing the accuracy of change points for the different treatments. Therefore, the calculation considers all theoretically possible change points resulting from the combinations of different spatial statistics ($n = 7$), bands/spectral indices (Sentinel-1: $n = 12$, Sentinel-2: $n = 129$), change point types ($n = 2$) and number of individual patches per treatment ($n = 3$, except for “DW”: $n = 18$). As a result, all calculated metrics based on all bands/spectral indices and spatial statistics combinations are taken into account to analyze the general potential of Sentinel-1 and Sentinel-2 to assess the different BETA-FOR treatments. For both Sentinel-1 and Sentinel-2, the great majority of accurately assessed change points are present in aggregated treatments (Fig. 9). Sentinel-1 derived metrics identified aggregated logs (AL) and aggregated total tree removal (AR) with highest percentage values (about 4% and 4.8%, respectively). Metrics based on Sentinel-2 determined accurate change points best in aggregated treatments of stumps (AW, about 4%), logs (AL, about 3.8%) and crowns (AK, about 5.1%). Sentinel-2 derived metrics could further identify accurate change points for the treatments of aggregated habitat trees (AH, about 0.8%) and the distributed treatments of stumps (DW, 0.1%), snags (DS, about 0.5%), logs and snags (DB, about 0.1%) and crowns (DK, about 0.3%). Aforementioned treatment implementations could not be accurately assessed based on Sentinel-1 metrics. Both Sentinel-1 and Sentinel-2 metrics did not determine any change point in the control treatment patches (CC).

(A) Sentinel-1



(B) Sentinel-2

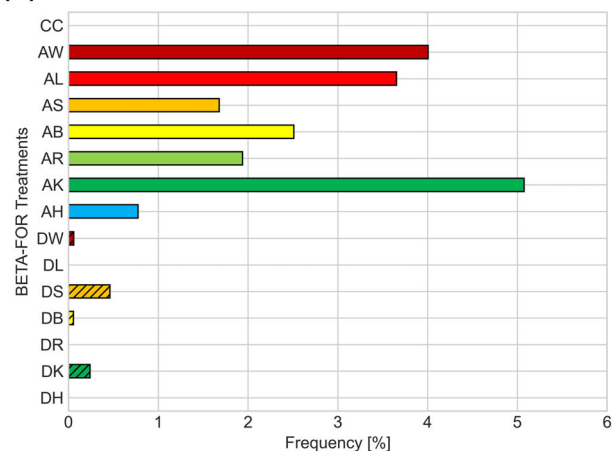


Figure 9. Comparative statistics of Sentinel-1 (A) and Sentinel-2 (B) metrics (combination of index and spatial statistics) assessing accurate change points grouped by treatment. Percentage values on the x-axis are calculated as grouped statistics per individual treatment type accounting for all metrics (Sentinel-1: $n = 84$, Sentinel-2: $n = 903$) and change point types ($n = 2$). The color-coding is aligned to Figure 1A with aggregated treatments as solid colors and distributed treatments with dashed bars.

Discussion

The accurate assessment of aggregated treatment implementations has been demonstrated for both Sentinel-1 and Sentinel-2 metrics. The finding that this was not possible for distributed treatments, confirmed our research hypothesis on the improved time-series analysis of forest structure complexities measured at pixel-level (aggregated treatments) opposing to sub-pixel dynamics (distributed treatments). The valid identification of treatment implementations for various aggregated structural complexities suggests that post-disturbance structures, such as complex vertical structures (snags, logs and snags, habitat trees) can be identified, in addition to downed (logs, crowns) or no deadwood (stumps, total tree removal). Therefore, recently published binary disturbance products (disturbed vs. undisturbed) in temperate forests (Senf et al., 2017; Thonfeld et al., 2022) could further investigate the mapping of standing deadwood. Nevertheless, the comprehensive testing from a standardized catalog of spectral indices (Montero et al., 2023) emphasizes that the selection of spectral indices and spatial statistics are key criteria. Most experimental treatments were assessed through heterogeneity metrics (e.g. coefficient of variation) and specific bands/spectral indices (Sentinel-1: VH polarization, Sentinel-2: NMDI). The potential of Sentinel-1 VH has previously been demonstrated in the context of tropical deforestation alerts (Reiche et al., 2021), biodiversity monitoring (Bae et al., 2019), forest structure modeling (Kacic et al., 2023) and the detection of drought affected forest stands (Kaiser et al., 2022). In the initial publication of the NMDI by Wang and Qu (2007), the spectral index combining near and short-wave infrared bands was proposed for the analysis of soil and vegetation moisture. Based on our time-series analysis, the NMDI can also be applied for the assessment of forest structure complexity analysis since the current applicability ranges from the evaluation of fire risk in different forest types (boreal forests: Akther and Hassan (2011); Chowdhury and Hassan (2015), arid regions: Wang et al. (2008), tropical forests: Babu et al. (2017)), over vegetation cover monitoring (de Oliveira Santos et al., 2021; Korhonen et al., 2017) to the downscaling of land surface temperature in arid areas (Yang et al., 2017). The varying consistency of remotely sensed metrics based on Sentinel-1 and Sentinel-2 highlights that there is no single-best metric assessing the different treatments, which is why a combination of complementary bands/spectral indices is recommended (Zeng et al., 2022).

Due to atmospheric artifacts, the number of time-series observations from Sentinel-2 was lower than for Sentinel-1 (Table 1). Nevertheless, the reduced actual analysis-ready temporal resolution of Sentinel-2 was not a clear drawback since Sentinel-2 derived metrics reach similar percentage values for the accurate detection of change

points characterizing treatment implementations (Fig. 9). In addition, the reduced number of observations for Sentinel-2 might be compensated by the high number of spectral bands, thus enabling the calculation of more and diverse spectral indices (Frampton et al., 2013; Montero et al., 2023). Furthermore, the potential measurements of vegetation shadowing (e.g. aggregated treatments) in optical spectra can be seen as a proxy for light structure conditions. The contrasting findings for the assessed treatment implementation dates for Sentinel-1 and Sentinel-2 might be a result of both spectral and temporal resolution (Figure S1). The analysis on the integration of Sentinel-2 bands for spectral indices assessing accurate treatment implementations highlight that bands in the near infrared, short-wave infrared and red wavelength are critical for the identification of forest structure complexity dynamics (Figure S2).

Although the BETA-FOR experimental design follows standardized conditions in terms of patch size, treatment types, land use history and forest composition, external influences such as natural disturbances can never be fully excluded. Based on the time-series analysis of Sentinel-1 and Sentinel-2, no change points outside the treatment implementation period were detected that match the change point characteristics of treatment implementations, thus generally suggesting the absence of major disturbance events, i.e. the presence of consistent environmental conditions. Furthermore, the criteria that treatment implementation change points need to reach maximum probability for the full time-series of a respective metric served as a further controlling factor of other disturbance influences: for more than 95% of all change points detecting treatment implementation events, the change point probability amounted to the maximum, thus confirming that the treatment implementation in November and December 2018 was the strongest change in forest structure during the full time-series available.

The vegetation dynamics of European temperate forests in recent years are specifically influenced by heatwaves and drought (Buras et al., 2020; Rakovec et al., 2022; Schuldt et al., 2020), resulting in disturbances (Senf et al., 2021; Senf & Seidl, 2021), but also long-term greening (Ghaderpour et al., 2023). Increasing disturbance frequencies at exceptional spatio-temporal scales have been reported by multiple studies (Senf et al., 2021; Senf & Seidl, 2021; Thonfeld et al., 2022). Additional dynamics of greening and warming in Italy have been extensively outlined by Ghaderpour et al. (2023) utilizing various MODIS products in combination with data on topography and precipitation. Overall, the study concludes that within the past decades most parts of Italy were characterized by greening and gradual warming, both being further controlling factors of future vegetation composition and health (Ghaderpour et al., 2023).

The present methodological approach investigating various forest structure complexities at local scale can be implemented at larger-scale by applying lessons-learned in terms of the capability of Sentinel-1 and Sentinel-2 to detect various aggregated structural complexities (minimum size: 50×50 m) based on certain spectral indices and spatial statistics. More specifically, the BEAST algorithm (Zhao et al., 2019) can be integrated for the decomposition of time-series into additive seasonal and trend components and the identification of change points with probability scores as moving window approach (e.g. at the spatial scale of BETA-FOR patches). Therefore, corresponding spatio-temporal signatures to BETA-FOR treatments can be identified as proxy for specific structural complexities (light-deadwood arrangements). For the analysis of recent dynamics in forests, the high spatio-temporal observations from Sentinel-1 and Sentinel-2 might be specifically suited, but also long time-series of Landsat enable the monitoring of forest structure for more than 40 years (Banskota et al., 2014; Irons et al., 2012).

The continuous monitoring of forest structure complexity in times of global climate warming, biodiversity loss and increasing disturbances is an essential task for supporting the future resilience of forests. Therefore, multiple research groups call for increased biodiversity conservation (Daz et al., 2020) and socio-ecological forest management (Müller et al., 2018; Nikinmaa et al., 2023). In addition, the correlation of remotely sensed time-series with in-situ data on forest biodiversity for spatio-temporal modeling of taxa and ecosystem functions should be further investigated to better link not only local to spaceborne remote sensing perspectives (Storch et al., 2023) but also different research disciplines, such as ecology and remote sensing (Cavender-Bares et al., 2022). The trans-disciplinary nature of the BETA-FOR project ranging from terrestrial forest structure measurements, over spaceborne remote sensing, to the comprehensive in-situ measurement of various taxa (e.g. microbes, plants, bats, birds, mammals) and ecosystem functions (e.g. soil diversity, microclimate, decomposition processes, above- and below-ground biomass) strongly supports the correlation of various data sources (Mueller, Mitesser, et al., 2022; Mueller, Scherer-Lorenzen, et al., 2022). Further analysis could investigate the relationship of terrestrial and spaceborne measures of forest structure complexity in order to bridge scales and subsequently model multi-taxa indicators and ecosystem functions.

Conclusion

In recent years, German forests are heavily affected by climate change-induced drought conditions (Buras et al., 2020; Rakovec et al., 2022; Schuldt et al., 2020). Specifically spruce

mono-cultures are characterized by large-scale canopy cover loss (Thonfeld et al., 2022), but also other coniferous and deciduous species negatively respond (e.g. crown defoliation) to increasing disturbances (Federal Ministry of Food and Agriculture (BMEL), 2021a, b). Because of the general mono-structure of German forests in terms of age, species and three-dimensional properties, adaptive management is needed to ensure future health and resilience (Müller et al., 2018). Within the framework of the BETA-FOR project, experimental silvicultural treatments were implemented to enhance structural complexity in six different regions of Germany (Mueller, Mitesser, et al., 2022; Mueller, Scherer-Lorenzen, et al., 2022). The present study focuses on a broad-leaved forest called “University Forest” in central Germany in which a patch network of aggregated and distributed treatments (diversification of light structures) was implemented in winter 2018. The deadwood structures range from total tree removal, stumps, logs and snags (standing deadwood with a cut crown), to downed crowns, lying and standing deadwood and habitat trees. Based on remote sensing time-series from Sentinel-1 and Sentinel-2 various forest structure complexities were characterized. Both sensors can accurately detect experimental treatments enhancing structural diversification when specific spectral indices and spatial statistics are selected. More specifically, there is a clear advantage for the identification of aggregated treatments based on Sentinel-1 and Sentinel-2 metrics. Overall, Sentinel-1 and Sentinel-2 show similar performance to characterize aggregated post-disturbance structures comprising not only no or downed deadwood but also standing deadwood. Nevertheless, Sentinel-1 might be beneficial when assessing forest structure complexity in areas of more persistent cloud coverage compared to our study area due to its insensitivity towards atmospheric conditions. The following bullet points serve as a summary of the main findings:

- The bands/indices derived from Sentinel-1 accurately assessed most treatment implementations are VH and VVVHP.
- Sentinel-2-based spectral indices which most accurately identified treatment implementations are NMDI and MBWI.
- Heterogeneity statistics (e.g. coefficient of variation) better assessed treatment implementations than extreme or average statistics for both Sentinel-1 and Sentinel-2.
- Accurate treatment implementations could generally be better identified in the trend component (Sentinel-1: 97.0%, Sentinel-2: 83.2%) but Sentinel-2 was capable to identify change points in the seasonality component (16.8%).
- Most of the accurate seasonality change points identifying treatment implementations were also assessed as trend change points (temporal overlap) for Sentinel-1 (66.6%) and Sentinel-2 (65.6%).

- The BEAST algorithm clearly identified the implementation of aggregated BETA-FOR treatments as strongest change in forest structure and differentiated this dominant disturbance event from minor gradients of change (e.g. natural dynamics) through the assessment of change point probabilities.

Acknowledgements

All authors on the paper have seen and approved the submitted version of the manuscript. Furthermore, all authors have substantially contributed to the work, and all persons entitled to co-authorship have been included. The manuscript has been submitted solely to Remote Sensing in Ecology and Conservation and it has not been published elsewhere, either in part or whole, nor is it in press or under consideration for publication in another journal. PK acknowledges funding from the DFG (Deutsche Forschungsgemeinschaft) within the framework of the Research Unit BETA-FOR (Enhancing the structural diversity between patches for improving multidiversity and multifunctionality in production forests) (grant no. FOR 5375/1, project number 459717468). FT acknowledges funding from the ForstEO project (FKZ 2220WK81A4), funded by the Federal Ministry of Food and Agriculture and Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection based on a decision of the German Bundestag from Waldklimafonds through the Fachagentur Nachwachsende Rohstoffe e.V. (FNR) Open Access funding enabled and organized by Projekt DEAL.

Conflict of Interest

The authors have no conflict of interest to declare.

Data Accessibility Statement

The Sentinel-1 and Sentinel-2 time-series data and statistical results produced in this study are available from the corresponding author upon request.

References

- Akther, M.S. & Hassan, Q.K. (2011) Remote sensing-based assessment of fire danger conditions over boreal forest. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, **4**(4), 992–999. Available from: <https://doi.org/10.1109/JSTARS.2011.2165940>
- Awty-Carroll, K., Bunting, P., Hardy, A. & Bell, G. (2019) An evaluation and comparison of four dense time series change detection methods using simulated data. *Remote Sensing*, **11** (23), 2779. Available from: <https://doi.org/10.3390/rs11232779>
- Babu, K. S., Vanama, V. S. K., Roy, A., & Prasad, P. R. (2017). Assessment of forest fire danger using automatic weather stations and modis terra satellite datasets for the State Madhya Pradesh, India. (pp. 1876–1881)., <https://doi.org/10.1109/ICACCI.2017.8126118>.
- Bae, S., Levick, S.R., Heidrich, L., Magdon, P., Leutner, B.F., Wöllauer, S. et al. (2019) Radar vision in the mapping of forest biodiversity from space. *Nature Communications*, **10** (1), 4757. Available from: <https://doi.org/10.1038/s41467-019-12737-x>
- Banskota, A., Kayastha, N., Falkowski, M.J., Wulder, M.A., Froese, R.E. & White, J.C. (2014) Forest monitoring using landsat time series data: a review. *Canadian Journal of Remote Sensing*, **40**(5), 362–384. Available from: <https://doi.org/10.1080/07038992.2014.987376>
- Berger, M., Moreno, J., Johannessen, J.A., Levelt, P.F. & Hanssen, R.F. (2012) ESA's sentinel missions in support of earth system science. *Remote Sensing of Environment*, **120**, 84–90. Available from: <https://doi.org/10.1016/j.rse.2011.07.023>
- Bhattarai, R., Rahimzadeh-Bajgiran, P., Weiskittel, A. & MacLean, D.A. (2020) Sentinel-2 based prediction of spruce budworm defoliation using red-edge spectral vegetation indices. *Remote Sensing Letters*, **11**(8), 777–786. Available from: <https://doi.org/10.1080/2150704X.2020.1767824>
- Buras, A., Rammig, A. & Zang, C.S. (2020) Quantifying impacts of the 2018 drought on european ecosystems in comparison to 2003. *Biogeosciences*, **17**(6), 1655–1672. Available from: <https://doi.org/10.5194/bg-17-1655-2020>
- Cai, Y., Liu, S. & Lin, H. (2020) Monitoring the vegetation dynamics in the dongting lake wetland from 2000 to 2019 using the beast algorithm based on dense landsat time series. *Applied Sciences*, **10**(12), 4209.
- Camarretta, N., Harrison, P.A., Bailey, T., Potts, B., Lucieer, A., Davidson, N. et al. (2020) Monitoring forest structure to guide adaptive management of forest restoration: a review of remote sensing approaches. *New Forests*, **51**, 573–596. Available from: <https://doi.org/10.1007/s11056-019-09754-5>
- Cavender-Bares, J., Schneider, F.D., Santos, M.J., Armstrong, A., Carnaval, A., Dahlin, K.M. et al. (2022) Integrating remote sensing with ecology and evolution to advance biodiversity conservation. *Nature Ecology & Evolution*, **6**(5), 506–519. Available from: <https://doi.org/10.1038/s41559-022-01702-5>
- Chaudhary, A., Burivalova, Z., Koh, L.P. & Hellweg, S. (2016) Impact of forest management on species richness: global meta-analysis and economic trade-offs. *Scientific Reports*, **6** (1), 23954. Available from: <https://doi.org/10.1038/srep23954>
- Chowdhury, E.H. & Hassan, Q.K. (2015) Development of a new daily-scale forest fire danger forecasting system using remote sensing data. *Remote Sensing*, **7**(3), 2431–2448. Available from: <https://doi.org/10.3390/rs70302431>
- Crabbe, R.A., Dash, J., Rodriguez-Galiano, V.F., Janous, D., Pavelka, M. & Marek, M.V. (2016) Extreme warm

- temperatures alter forest phenology and productivity in europe. *Science of the Total Environment*, **563**, 486–495.
- Daz, S., Zafra-Calvo, N., Purvis, A., Verburg, P.H., Obura, D., Leadley, P. et al. (2020) Set ambitious goals for biodiversity and sustainability. *Science*, **370**(6515), 411–413. Available from: <https://doi.org/10.1126/science.abe1530>
- de Oliveira Santos, R., Delgado, R.C., Vilanova, R.S., de Santana, R.O., de Andrade, C.F., Teodoro, P.E. et al. (2021) NMDI application for monitoring different vegetation covers in the atlantic forest biome, Brazil. *Weather and Climate Extremes*, **33**, 100329. Available from: <https://doi.org/10.1016/j.wace.2021.100329>
- Delegido, J., Verrelst, J., Alonso, L. & Moreno, J. (2011) Evaluation of sentinel-2 red-edge bands for empirical estimation of green lai and chlorophyll content. *Sensors*, **11** (7), 7063–7081. Available from: <https://doi.org/10.3390/s110707063>
- Dieler, J., Uhl, E., Biber, P., Müller, J., Rötzer, T. & Pretzsch, H. (2017) Effect of forest stand management on species composition, structural diversity, and productivity in the temperate zone of europe. *European Journal of Forest Research*, **136**(4), 739–766. Available from: <https://doi.org/10.1007/s10342-017-1056-1>
- Dormann, C.F., Bagnara, M., Boch, S., Hinderling, J., Janeiro-Otero, A., Schäfer, D. et al. (2020) Plant species richness increases with light availability, but not variability, in temperate forests understorey. *BMC Ecology*, **20**(1), 43. Available from: <https://doi.org/10.1186/s12898-020-00311-9>
- European Commission (2020a). Eu biodiversity strategy for 2030. Available from: https://environment.ec.europa.eu/strategy/biodiversity-strategy-2030_en [Accessed 20 July 2023].
- European Commission (2020b). Eu forest strategy for 2030. Available from: https://environment.ec.europa.eu/strategy/forest-strategy_en#:~:text=New%20EU%20forest%20strategy%20for%202030&text=The%20strategy%20will%20contribute%20to,and%20climate%20neutrality%20by%202050 [Accessed 20 July 2023].
- European Commission (2023). Nature restoration law. Available from: https://environment.ec.europa.eu/topics/nature-and-biodiversity/nature-restoration-law_en. [Accessed 20 July 2023].
- Federal Ministry of Food and Agriculture (BMEL). Ergebnisse der waldzustandserhebung. 2021a. Available from: https://www.bmel.de/SharedDocs/Downloads/DE/Broschueren/ergebnisse-waldzustandserhebung-2021.pdf?__blob=publicationFile&v=10. [Accessed 04 September 2023].
- Federal Ministry of Food and Agriculture (BMEL). Waldbericht der bundesregierung. 2021b. Available from: https://www.bmel.de/SharedDocs/Downloads/DE/Broschueren/waldbericht2021.pdf?__blob=publicationFile&v=11. [Accessed 04 September 2023].
- Frampton, W.J., Dash, J., Watmough, G. & Milton, E.J. (2013) Estimation of biophysical variables in vegetation. *ISPRS Journal of Photogrammetry and Remote Sensing*, **82**, 83–92. Available from: <https://doi.org/10.1016/j.isprsjprs.2013.04.007>
- Fu, Y.H., Geng, X., Chen, S., Wu, H., Hao, F., Zhang, X. et al. (2023) Global warming is increasing the discrepancy between green (actual) and thermal (potential) seasons of temperate trees. *Global Change Biology*, **29**(5), 1377–1389.
- Gao, B.-C. (1996) NDWI—a normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment*, **58**(3), 257–266. Available from: [https://doi.org/10.1016/S0034-4257\(96\)00067-3](https://doi.org/10.1016/S0034-4257(96)00067-3), <https://www.sciencedirect.com/science/article/pii/S0034425796000673>
- Ghaderpour, E., Mazzanti, P., Mugnozza, G.S. & Bozzano, F. (2023) Coherency and phase delay analyses between land cover and climate across Italy via the least-squares wavelet software. *International Journal of Applied Earth Observation and Geoinformation*, **118**, 103241.
- Ghaderpour, E. & Vujadinovic, T. (2020) Change detection within remotely sensed satellite image time series via spectral analysis. *Remote Sensing*, **12**(23), 4001. Available from: <https://doi.org/10.3390/rs12234001>
- Gonzalez, A., Vihervaara, P., Balvanera, P., Bates, A.E., Bayraktarov, E., Bellingham, P.J. et al. (2023) A global biodiversity observing system to unite monitoring and guide action. *Nature Ecology & Evolution*, **7**, 1947–1952. Available from: <https://doi.org/10.1038/s41559-023-02171-0>
- Google Earth Engine Developers (2022). Sentinel-2 cloud masking with s2cloudless. Available from: <https://github.com/google/earthengine-community/blob/master/tutorials/sentinel-2-s2cloudless/index.ipynb>. [Accessed 10 October 2022].
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. & Moore, R. (2017) Google earth engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, **202**, 18–27. Available from: <https://doi.org/10.1016/j.rse.2017.06.031>
- Großmann, J., Carlson, L., Kändler, G., Pyttel, P., Kleinschmit, J.R.G. & Bauhus, J. (2023) Evaluating retention forestry 10 years after its introduction in temperate forests regarding the provision of tree-related microhabitats and dead wood. *European Journal of Forest Research*, **142**, 1125–1147. Available from: <https://doi.org/10.1007/s10342-023-01581-w>
- Hajek, P., Link, R.M., Nock, C.A., Bauhus, J., Gebauer, T., Gessler, A. et al. (2022) Mutually inclusive mechanisms of drought-induced tree mortality. *Global Change Biology*, **28** (10), 3365–3378. Available from: <https://doi.org/10.1111/gcb.16146>
- Halme, P., Toivanen, T., Honkanen, M., Kotiaho, J.S., Mönkkönen, M. & Timonen, J. (2010) Flawed meta-analysis of biodiversity effects of forest management. *Conservation Biology*, **24**(4), 1154–1156. Available from: <https://doi.org/10.1111/j.1523-1739.2010.01542.x>

- Heidrich, L., Bae, S., Levick, S., Seibold, S., Weisser, W., Krzystek, P. et al. (2020) Heterogeneity–diversity relationships differ between and within trophic levels in temperate forests. *Nature Ecology & Evolution*, **4**(9), 1204–1212. Available from: <https://doi.org/10.1038/s41559-020-1245-z>
- Hoffmann, J., Muro, J. & Dubovyk, O. (2022) Predicting species and structural diversity of temperate forests with satellite remote sensing and deep learning. *Remote Sensing*, **14**(7), 1631. Available from: <https://doi.org/10.3390/rs14071631>
- Holzwarth, S., Thonfeld, F., Abdullahi, S., Asam, S., da Ponte Canova, E., Gessner, U. et al. (2020) Earth observation based monitoring of forests in Germany: a review. *Remote Sensing*, **12**(21), 3570. Available from: <https://doi.org/10.3390/rs12213570>
- Holzwarth, S., Thonfeld, F., Kacic, P., Abdullahi, S., Asam, S., Coleman, K. et al. (2023) Earth-observation-based monitoring of forests in Germany—recent progress and research frontiers: a review. *Remote Sensing*, **15**(17), 4234. Available from: <https://doi.org/10.3390/rs15174234>
- Huete, A., Didan, K., Miura, T., Rodriguez, E.P., Gao, X. & Ferreira, L.G. (2002) Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment*, **83**(1–2), 195–213. Available from: [https://doi.org/10.1016/S0034-4257\(02\)00096-2](https://doi.org/10.1016/S0034-4257(02)00096-2)
- Huete, A.R. (2012) Vegetation indices, remote sensing and forest monitoring. *Geography Compass*, **6**(9), 513–532. Available from: <https://doi.org/10.1111/j.1749-8198.2012.00507.x>
- Irons, J.R., Dwyer, J.L. & Barsi, J.A. (2012) The next landsat satellite: the landsat data continuity mission. *Remote Sensing of Environment*, **122**, 11–21. Available from: <https://doi.org/10.1016/j.rse.2011.08.026>
- Jakobsson, R., Olofsson, E. & Ambrose-Oji, B. (2020) Stakeholder perceptions, management and impacts of forestry conflicts in southern Sweden. *Scandinavian Journal of Forest Research*, **36**(1), 68–82. Available from: <https://doi.org/10.1080/02827581.2020.1854341>
- Jetz, W., Cavender-Bares, J., Pavlick, R., Schimel, D., Davis, F.W., Asner, G.P. et al. (2016) Monitoring plant functional diversity from space. *Nature Plants*, **2**(3), 16024. Available from: <https://doi.org/10.1038/nplants.2016.24>
- Kacic, P., Thonfeld, F., Gessner, U. & Kuenzer, C. (2023) Forest structure characterization in Germany: novel products and analysis based on GEDI, Sentinel-1 and Sentinel-2 data. *Remote Sensing*, **15**(8), 1969. Available from: <https://doi.org/10.3390/rs15081969>
- Kaiser, P., Buddenbaum, H., Nink, S. & Hill, J. (2022) Potential of sentinel-1 data for spatially and temporally high-resolution detection of drought affected forest stands. *Forests*, **13**(12), 2148. Available from: <https://doi.org/10.3390/f13122148>
- Kennedy, R.E., Yang, Z. & Cohen, W.B. (2010) Detecting trends in forest disturbance and recovery using yearly landsat time series: 1. Landtrendr—temporal segmentation algorithms. *Remote Sensing of Environment*, **114**(12), 2897–2910.
- König, S., Thonfeld, F., Förster, M., Dubovyk, O. & Heurich, M. (2023) Assessing combinations of landsat, sentinel-2 and sentinel-1 time series for detecting bark beetle infestations. *GIScience & Remote Sensing*, **60**(1), 2226515. Available from: <https://doi.org/10.1080/15481603.2023.2226515>
- Korhonen, L., Packalen, P. & Rautiainen, M. (2017) Comparison of sentinel-2 and landsat 8 in the estimation of boreal forest canopy cover and leaf area index. *Remote Sensing of Environment*, **195**, 259–274. Available from: <https://doi.org/10.1016/j.rse.2017.03.021>
- Krumm, F., Bauhus, J., Larsen, J.B., Knoke, T., Poetzelsberger, E., Schuck, A. et al. (2023) «Closer-to-nature forest management»: was ist neu an diesem konzept? *Schweizerische Zeitschrift für Forstwesen*, **174**(3), 158–161. Available from: <https://doi.org/10.3188/szf.2023.0158>
- Li, J., Li, Z.-L., Wu, H. & You, N. (2022) Trend, seasonality, and abrupt change detection method for land surface temperature time-series analysis: evaluation and improvement. *Remote Sensing of Environment*, **280**, 113222.
- Li, W., Guo, W.Y., Pasgaard, M., Niu, Z., Wang, L., Chen, F. et al. (2023) Human fingerprint on structural density of forests globally. *Nature Sustainability*, **6**, 368–379. Available from: <https://doi.org/10.1038/s41893-022-01020-5>
- Liu, X., Frey, J., Munteanu, C., Still, N. & Koch, B. (2023) Mapping tree species diversity in temperate montane forests using sentinel-1 and sentinel-2 imagery and topography data. *Remote Sensing of Environment*, **292**, 113576. Available from: <https://doi.org/10.1016/j.rse.2023.113576>
- Main-Knorn, M., Pflug, B., Louis, J., Debaecker, V., Müller-Wilm, U. & Gascon, F. (2017) Sen2cor for sentinel-2. <https://doi.org/10.1117/12.2278218>
- Malenovsky, Z., Rott, H., Cihlar, J., Schaepman, M.E., García-Santos, G., Fernandes, R. et al. (2012) Sentinels for science: potential of sentinel-1, -2, and -3 missions for scientific observations of ocean, cryosphere, and land. *Remote Sensing of Environment*, **120**, 91–101. Available from: <https://doi.org/10.1016/j.rse.2011.09.026>
- Messier, C., Bauhus, J., Sousa-Silva, R., Auge, H., Baeten, L., Barsoum, N. et al. (2021) For the sake of resilience and multifunctionality, let's diversify planted forests! *Conservation Letters*, **15**(1), e12829. Available from: <https://doi.org/10.1111/conl.12829>
- Montero, D., Aybar, C., Mahecha, M.D., Martinuzzi, F., Soechting, M. & Wieneke, S. (2023) A standardized catalogue of spectral indices to advance the use of remote sensing in earth system research. *Scientific Data*, **10**(1), 197. Available from: <https://doi.org/10.1038/s41597-023-02096-0>
- Mueller, J., Mitesser, O., Cadotte, M.W., van der Plas, F., Mori, A.S., Ammer, C. et al. (2022) Enhancing the structural diversity between forest patches—a concept and real-world experiment to study biodiversity,

- multifunctionality and forest resilience across spatial scales. *Global Change Biology*, **29**(6), 1437–1450. Available from: <https://doi.org/10.1111/gcb.16564>
- Mueller, J., Scherer-Lorenzen, M., Ammer, C., Eisenhauer, N., Seidel, D., Schuldt, B. et al. (2022) BETA-FOR: Erhöhung der strukturellen Diversität zwischen Waldbeständen zur Erhöhung der Multidiversität und Multifunktionalität in Produktionswäldern. Antragstext für die DFG Forschungsgruppe FOR 5375. Technical report <https://opus.bibliothek.uni-wuerzburg.de/frontdoor/index/index/docId/29084>
- Müller, J., Noss, R.F., Thorn, S., Bässler, C., Leverkus, A.B. & Lindenmayer, D. (2018) Increasing disturbance demands new policies to conserve intact forest. *Conservation Letters*, **12**(1), e12449. Available from: <https://doi.org/10.1111/conl.12449>
- Mullissa, A., Vollrath, A., Odongo-Braun, C., Slagter, B., Balling, J., Gou, Y. et al. (2021) Sentinel-1 SAR backscatter analysis ready data preparation in google earth engine. *Remote Sensing*, **13**(10), 1954. Available from: <https://doi.org/10.3390/rs13101954>
- Muys, B., Angelstam, P., Bauhus, J., Bouriaud, L., Jactel, H., Kraigher, H. et al. (2022) Forest biodiversity in Europe. <https://doi.org/10.36333/fs13>
- Nabuurs, G.-J., Harris, N., Sheil, D., Palahi, M., Chirici, G., Boissière, M. et al. (2022) Glasgow forest declaration needs new modes of data ownership. *Nature Climate Change*, **12**(5), 415–417. Available from: <https://doi.org/10.1038/s41558-022-01343-3>
- Nikinmaa, L., Lindner, M., Cantarello, E., Gardiner, B., Jacobsen, J.B., Jump, A.S. et al. (2023) A balancing act: principles, criteria and indicator framework to operationalize social-ecological resilience of forests. *Journal of Environmental Management*, **331**, 117039. Available from: <https://doi.org/10.1016/j.jenvman.2022.117039>
- Paillet, Y., Bergès, L., Hjältén, J., Ódor, P., Avon, C., Bernhardt-Römermann, M.A. et al. (2010) Biodiversity differences between managed and unmanaged forests: meta-analysis of species richness in Europe. *Conservation Biology*, **24**(1), 101–112. Available from: <https://doi.org/10.1111/j.1523-1739.2009.01399.x>
- Penone, C., Allan, E., Soliveres, S., Felipe-Lucia, M.R., Gossner, M.M., Seibold, S. et al. (2018) Specialisation and diversity of multiple trophic groups are promoted by different forest features. *Ecology Letters*, **22**(1), 170–180. Available from: <https://doi.org/10.1111/ele.13182>
- Popkin, G. (2021) Forest fight. *Science*, **374**(6572), 1184–1189. Available from: <https://www.science.org/doi/pdf/10.1126/science.acx9733>
- Rakovec, O., Samaniego, L., Hari, V., Markonis, Y., Moravec, V., Thober, S. et al. (2022) The 2018–2020 multi-year drought sets a new benchmark in Europe. *Earth's Future*, **10**(3), e2021EF002394. Available from: <https://doi.org/10.1029/2021ef002394>
- Reiche, J., Mullissa, A., Slagter, B., Gou, Y., Tsendbazar, N.E., Odongo-Braun, C. et al. (2021) Forest disturbance alerts for The Congo basin using sentinel-1. *Environmental Research Letters*, **16**(2), 024005. Available from: <https://doi.org/10.1088/1748-9326/abd0a8>
- Rothacher, J., Hagge, J., Bässler, C., Brandl, R., Gruppe, A. & Müller, J. (2023) Logging operations creating snags, logs, and stumps under open and closed canopies promote stand-scale beetle diversity. *Forest Ecology and Management*, **540**, 121022. Available from: <https://doi.org/10.1016/j.foreco.2023.121022>
- Rouse, J.W., Haas, R.H., Schell, J.A. & Deering, D.W. (1974) Monitoring vegetation systems in the great plains with erts. *NASA Special Publications*, **351**(1), 309.
- Sandström, J., Bernes, C., Junninen, K., Lohmus, A., Macdonald, E., Müller, J. et al. (2019) Impacts of dead wood manipulation on the biodiversity of temperate and boreal forests. A systematic review. *Journal of Applied Ecology*, **56**(7), 1770–1781. Available from: <https://doi.org/10.1111/1365-2664.13395>
- Scharvogel, D., Brandmeier, M. & Weis, M. (2020) A deep learning approach for calamity assessment using sentinel-2 data. *Forests*, **11**(12), 1239. Available from: <https://doi.org/10.3390/f11121239>
- Schuldt, B., Buras, A., Arend, M., Vitasse, Y., Beierkuhnlein, C., Damm, A. et al. (2020) A first assessment of the impact of the extreme 2018 summer drought on central European forests. *Basic and Applied Ecology*, **45**, 86–103. Available from: <https://doi.org/10.1016/j.baae.2020.04.003>
- Schulze, E.-D. & Ammer, C. (2015) Spannungsfeld forstwirtschaft und naturschutz. *Biologie in unserer Zeit*, **45**(5), 304–314. Available from: <https://doi.org/10.1002/biuz.201510574>
- Senf, C., Buras, A., Zang, C.S., Rammig, A. & Seidl, R. (2020) Excess forest mortality is consistently linked to drought across Europe. *Nature Communications*, **11**(1), 6200. Available from: <https://doi.org/10.1038/s41467-020-19924-1>
- Senf, C., Pflugmacher, D., Hostert, P. & Seidl, R. (2017) Using landsat time series for characterizing forest disturbance dynamics in the coupled human and natural systems of central Europe. *ISPRS Journal of Photogrammetry and Remote Sensing*, **130**, 453–463. Available from: <https://doi.org/10.1016/j.isprsjprs.2017.07.004>
- Senf, C., Seibald, J. & Seidl, R. (2021) Increasing canopy mortality affects the future demographic structure of Europe's forests. *One Earth*, **4**(5), 749–755. Available from: <https://doi.org/10.1016/j.oneear.2021.04.008>
- Senf, C. & Seidl, R. (2021) Mapping the forest disturbance regimes of Europe. *Nature Sustainability*, **4**(1), 63–70. Available from: <https://doi.org/10.1038/s41893-020-00609-y>
- Skidmore, A.K., Coops, N.C., Neinavaz, E., Ali, A., Schaepman, M.E., Paganini, M. et al. (2021) Priority list of biodiversity metrics to observe from space. *Nature Ecology & Evolution*, **5**(7), 896–906. Available from: <https://doi.org/10.1038/s41559-021-01451-x>

- Storch, F., Boch, S., Gossner, M.M., Feldhaar, H., Ammer, C., Schall, P. et al. (2023) Linking structure and species richness to support forest biodiversity monitoring at large scales. *Annals of Forest Science*, **80**(1), 3. Available from: <https://doi.org/10.1186/s13595-022-01169-1>
- Sun, Y., Qin, Q., Ren, H., Zhang, T. & Chen, S. (2020) Red-edge band vegetation indices for leaf area index estimation from sentinel-2/msi imagery. *IEEE Transactions on Geoscience and Remote Sensing*, **58**(2), 826–840. Available from: <https://doi.org/10.1109/TGRS.2019.2940826>
- Teillet, P., Staenz, K. & Williams, D. (1997) Effects of spectral, spatial, and radiometric characteristics on remote sensing vegetation indices of forested regions. *Remote Sensing of Environment*, **61**(1), 139–149. Available from: [https://doi.org/10.1016/S0034-4257\(96\)00248-9](https://doi.org/10.1016/S0034-4257(96)00248-9)
- Thonfeld, F., Gessner, U., Holzwarth, S., Kriese, J., da Ponte, E., Huth, J. et al. (2022) A first assessment of canopy cover loss in Germany's forests after the 2018–2020 drought years. *Remote Sensing*, **14**(3), 562. Available from: <https://doi.org/10.3390/rs14030562>
- Van Der Plas, F., Manning, P., Allan, E., Scherer-Lorenzen, M., Verheyen, K., Wirth, C. et al. (2016) Jack-of-all-trades effects drive biodiversity–ecosystem multifunctionality relationships in european forests. *Nature Communications*, **7**(1), 11109. Available from: <https://doi.org/10.1038/ncomms11109>
- Van Der Plas, F., Manning, P., Soliveres, S., Allan, E., Scherer-Lorenzen, M., Verheyen, K. et al. (2016) Biotic homogenization can decrease landscape-scale forest multifunctionality. *Proceedings of the National Academy of Sciences of the United States of America*, **113**(13), 3557–3562. Available from: <https://doi.org/10.1073/pnas.1517903113>
- Verbesselt, J., Hyndman, R., Newnham, G. & Culvenor, D. (2010) Detecting trend and seasonal changes in satellite image time series. *Remote Sensing of Environment*, **114**(1), 106–115. Available from: <https://doi.org/10.1016/j.rse.2009.08.014>
- Wang, L. & Qu, J.J. (2007) NMDI: a normalized multi-band drought index for monitoring soil and vegetation moisture with satellite remote sensing. *Geophysical Research Letters*, **34**(20), L20405. Available from: <https://doi.org/10.1029/2007GL031021>
- Wang, L., Qu, J.J. & Hao, X. (2008) Forest fire detection using the normalized multi-band drought index (nm-di) with satellite measurements. *Agricultural and Forest Meteorology*, **148**(11), 1767–1776. Available from: <https://doi.org/10.1016/j.agrformet.2008.06.005>
- Xue, J. & Su, B. (2017) Significant remote sensing vegetation indices: a review of developments and applications. *Journal of Sensors*, **2017**, 1353691. Available from: <https://doi.org/10.1155/2017/1353691>
- Yang, Y., Cao, C., Pan, X., Li, X. & Zhu, X. (2017) Downscaling land surface temperature in an arid area by using multiple remote sensing indices with random forest regression. *Remote Sensing*, **9**(8), 789. Available from: <https://doi.org/10.3390/rs9080789>
- Zeller, L., Caicoya, A.T. & Pretzsch, H. (2021) Analyzing the effect of silvicultural management on the trade-off between stand structural heterogeneity and productivity over time. *European Journal of Forest Research*, **140**(3), 615–634. Available from: <https://doi.org/10.1007/s10342-020-01350-z>
- Zeller, L., Förster, A., Key, C., Meyer, P., Roschak, C. & Ammer, C. (2023) What does literature tell us about the relationship between forest structural attributes and species richness in temperate forests? – a review. *Ecological Indicators*, **153**, 110383. Available from: <https://doi.org/10.1016/j.ecolind.2023.110383>
- Zellweger, F., Coomes, D., Lenoir, J., Depauw, L., Maes, S.L., Wulf, M. et al. (2019) Seasonal drivers of understorey temperature buffering in temperate deciduous forests across europe. *Global Ecology and Biogeography*, **28**(12), 1774–1786. Available from: <https://doi.org/10.1111/geb.12991>
- Zellweger, F., de Frenne, P., Lenoir, J., Vangansbeke, P., Verheyen, K., Bernhardt-Römermann, M. et al. (2020) Forest microclimate dynamics drive plant responses to warming. *Science*, **368**(6492), 772–775. Available from: <https://doi.org/10.1126/science.aba6880>
- Zeng, Y., Hao, D., Huete, A., Dechant, B., Berry, J., Chen, J.M. et al. (2022) Optical vegetation indices for monitoring terrestrial ecosystems globally. *Nature Reviews Earth and Environment*, **3**(7), 477–493. Available from: <https://doi.org/10.1038/s43017-022-00298-5>
- Zhao, K., Wulder, M.A., Hu, T., Bright, R., Wu, Q., Qin, H. et al. (2019) Detecting change-point, trend, and seasonality in satellite time series data to track abrupt changes and nonlinear dynamics: a Bayesian ensemble algorithm. *Remote Sensing of Environment*, **232**, 111181. Available from: <https://doi.org/10.1016/j.rse.2019.04.034>
- Zhu, Z. & Woodcock, C.E. (2014) Continuous change detection and classification of land cover using all available landsat data. *Remote Sensing of Environment*, **144**, 152–171.
- Zimmermann, S. & Hoffmann, K. (2020) Evaluating the capabilities of sentinel-2 data for large-area detection of bark beetle infestation in the central german uplands. *Journal of Applied Remote Sensing*, **14**(2), 024515. Available from: <https://doi.org/10.1117/1.JRS.14.024515>

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1. Table of spectral indices calculated based on Montero et al. 2023 (<https://doi.org/10.1038/s41597-023-02096-0>).

Figure S1. Comparative statistics of Sentinel-1 (a) and Sentinel-2 (b) dates assessing accurate change points. The

x-axis values present the frequency of accurate change points assessed.

Figure S2. Analysis of Sentinel-2 bands contributing to the ten best spectral indices (according to Fig. 6) assessing accurate change points. The values on the x-axis indicate how often the various multispectral bands were included

in the calculation of those spectral indices that allowed for the detection of the silvicultural treatments. The band-wise calculation only considers the ten best spectral indices (according to Fig. 6) weighted (multiplied) by the number of accurate change points assessed per spectral index.